

AI-Powered Enterprise E-Commerce Platform

Feasibility Study Report

Document Information

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1. Introduction

Before investing time, money, and resources into a software project, it is important to determine whether the project is feasible.

A feasibility study evaluates whether a project can be successfully developed and whether it can achieve its intended objectives.

This document evaluates the feasibility of developing the AI-Powered Enterprise E-Commerce Platform.

2. Purpose

The purpose of this document is to:

- determine whether the project is practical,
 - identify potential challenges,
 - estimate costs and benefits,
 - evaluate technical requirements,
 - support decision making.
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3. Project Overview

The proposed system is an enterprise-level e-commerce platform integrated with Artificial Intelligence technologies.

The system aims to provide:

Traditional Features

- User Management
- Product Management
- Inventory Management
- Shopping Cart
- Order Processing
- Payment Processing

AI Features

- Recommendation System
- Fraud Detection
- Sales Forecasting
- Customer Segmentation
- AI Assistant

The system will use:

- Java (Spring Boot)
 - Python (FastAPI)
 - PostgreSQL
 - Docker
 - Machine Learning Technologies
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4. Need for the System

Modern businesses face several challenges.

Examples include:

- increasing customer expectations,
- growing competition,
- fraudulent activities,
- inventory management problems,
- lack of intelligent analytics.

Traditional e-commerce systems often lack advanced AI capabilities.

Therefore, there is a need for an intelligent enterprise platform that improves both customer experience and business decision-making.

5. Feasibility Analysis

Feasibility can be divided into several categories.

5.1 Technical Feasibility

Technical feasibility determines whether the required technology and skills are available.

Required Technologies

Backend Technologies

- Java
- Spring Boot
- REST APIs
- JWT Authentication

AI Technologies

- Python
- FastAPI
- Scikit-Learn
- XGBoost
- Pandas
- NumPy

Database Technologies

- PostgreSQL

- Redis
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DevOps Technologies

- Docker
 - Docker Compose
 - Git
 - GitHub
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Technical Availability

Most technologies are:

- open source,
- well documented,
- widely used in industry.

Large communities and documentation are available.

Technical Challenges

Possible challenges include:

Java and Python integration.

AI model deployment.

Scalability considerations.

Security implementation.

Microservice communication.

Technical Feasibility Conclusion

The project is considered:

Highly Feasible

because all required technologies are available and widely adopted.

5.2 Economic Feasibility

Economic feasibility evaluates whether the expected benefits justify the costs.

Estimated Development Costs

Hardware Costs

Item	Estimated Cost
Personal Computer	Existing Resource
Internet	Existing Resource
Cloud Resources	Optional

Software Costs

Software	Cost
Java	Free
Python	Free
PostgreSQL	Free
Docker	Free
GitHub	Free
VS Code	Free

Development Costs

For academic purposes:

Minimal.

For commercial deployment:

Cloud and infrastructure costs may apply.

Economic Benefits

Increased Sales

Personalized recommendations can increase sales.

Reduced Fraud Losses

Fraud detection can reduce financial losses.

Better Inventory Planning

Forecasting can reduce unnecessary costs.

Reduced Manual Work

Automation improves efficiency.

Economic Feasibility Conclusion

The project is:

Economically Feasible

because development costs are relatively low while potential benefits are high.

5.3 Operational Feasibility

Operational feasibility determines whether users will accept and use the system.

Customers

Expected to accept:

- personalized recommendations,
 - improved shopping experiences,
 - intelligent assistance.
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Administrators

Expected to benefit from:

- analytics,
 - reports,
 - fraud monitoring.
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Business Owners

Expected to benefit from:

- forecasting,
 - intelligent insights,
 - decision support.
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Operational Challenges

Possible challenges:

- user resistance to change,
 - learning new technologies,
 - trust in AI decisions.
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Operational Feasibility Conclusion

The project is considered:

Operationally Feasible

because it provides clear benefits to all stakeholders.

5.4 Schedule Feasibility

Schedule feasibility determines whether the project can be completed within a reasonable period.

Proposed Development Phases

Phase 1

Requirements Analysis

Phase 2

System Design

Phase 3

Database Design

Phase 4

Backend Development

Phase 5

AI Development

Phase 6

Frontend Development

Phase 7

Testing and Deployment

Estimated Timeline

Phase	Duration
Requirements	2 Weeks
Design	2 Weeks
Backend Development	5 Weeks
AI Development	4 Weeks
Frontend Development	4 Weeks
Testing	2 Weeks
Deployment	1 Week

Estimated Total:

4–5 Months

Schedule Risks

- Learning curve
 - AI model complexity
 - Integration difficulties
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Schedule Feasibility Conclusion

The project is considered:

Moderately Feasible

with proper planning and continuous development.

5.5 Legal Feasibility

Legal feasibility determines whether the system complies with legal requirements.

Important Considerations

User Privacy

Customer data should be protected.

Data Protection

Personal information should be stored securely.

Security Regulations

Appropriate security measures should be implemented.

Intellectual Property

Open-source licenses should be respected.

Legal Feasibility Conclusion

The project is considered:

Legally Feasible

provided that proper security and privacy measures are implemented.

6. Cost-Benefit Analysis

Costs

Development Time

Moderate.

Learning Effort

High.

Infrastructure Costs

Low to Moderate.

Benefits

Technical Benefits

- Enterprise experience
 - AI integration experience
 - Microservices experience
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Business Benefits

- Increased sales
 - Better decision making
 - Reduced fraud
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Educational Benefits

- Portfolio enhancement
 - Industry readiness
 - Improved employability
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7. Risks

Technical Risks

- Integration challenges
 - Performance issues
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Data Risks

- Insufficient datasets
 - Poor data quality
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Security Risks

- Cyber attacks
 - Data breaches
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Project Risks

- Requirement changes
 - Schedule delays
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8. Assumptions

The following assumptions have been made.

1. Development tools remain available.
2. Sufficient datasets can be obtained.
3. Internet access remains available.
4. Team members continue learning required technologies.

9. Recommendations

The following recommendations are proposed.

Use modular architecture.

Use microservices.

Begin with core functionalities.

Add AI features gradually.

Maintain detailed documentation.

Use version control continuously.

10. Feasibility Summary

Feasibility Type	Result
Technical	Highly Feasible
Economic	Feasible
Operational	Feasible
Schedule	Moderately Feasible
Legal	Feasible

11. Final Recommendation

Based on the analysis performed, the AI-Powered Enterprise E-Commerce Platform is considered:

FEASIBLE

for development.

The project provides:

- educational value,
- business value,
- portfolio value,
- and industry relevance.

Although certain risks exist, they can be managed through proper planning and incremental development.

12. Conclusion

The feasibility analysis indicates that the proposed project can be successfully developed using available technologies and resources.

The system is expected to:

- improve customer experiences,
- support business decision-making,
- provide intelligent analytics,
- and demonstrate modern enterprise software engineering practices.

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